

Malnutrition and its Oral Manifestations



Dr. Humaira Aman Ali



OBJECTIVES

At the end of the lecture, students would be able to:

- ❑ **Define and classify malnutrition.**
 - ❑ **Explain the causes of malnutrition.**
 - ❑ **Understand the impact of malnutrition on oral health.**
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MALNUTRITION

- ❑ "mal" is a prefix derived from Latin that means bad, abnormal, ill, or inadequate. It is typically used to denote a disorder, disease, or a negative condition.
- ❑ Nutrition is the study of nutrients, and the branch of science that involves nutrients and nutrition.
 - ❑ Nutrition is the biochemical and physiological process by which a living organism obtains, absorbs, and uses food and water to support life, growth, and tissue maintenance.
 - ❑ It involves the study of how the body uses nutrients—such as proteins, carbohydrates, fats, vitamins, and minerals—to survive, repair tissues, and prevent diseases.

DEFINITION

Deficiency

Excess

Imbalance

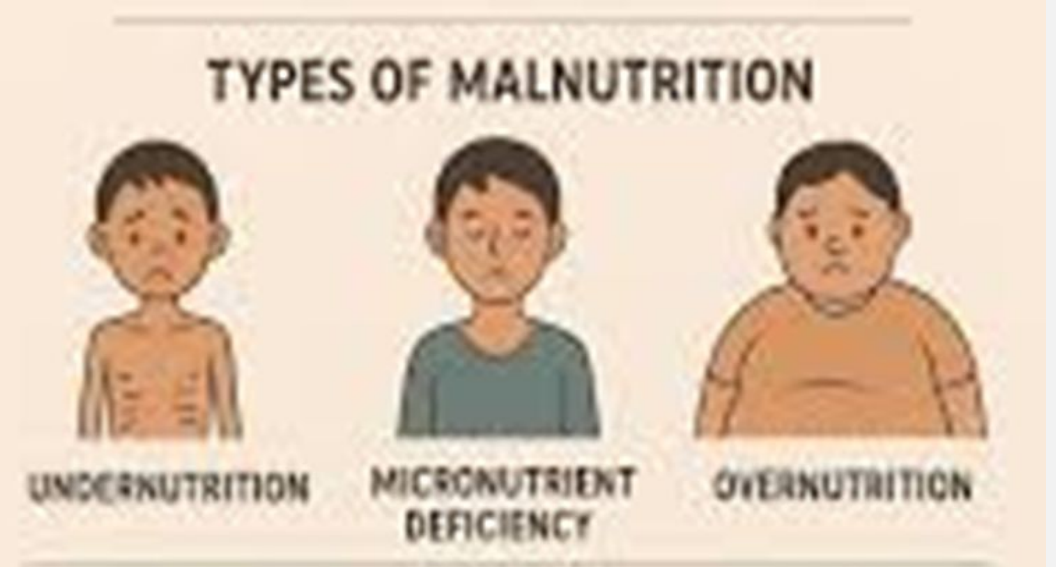
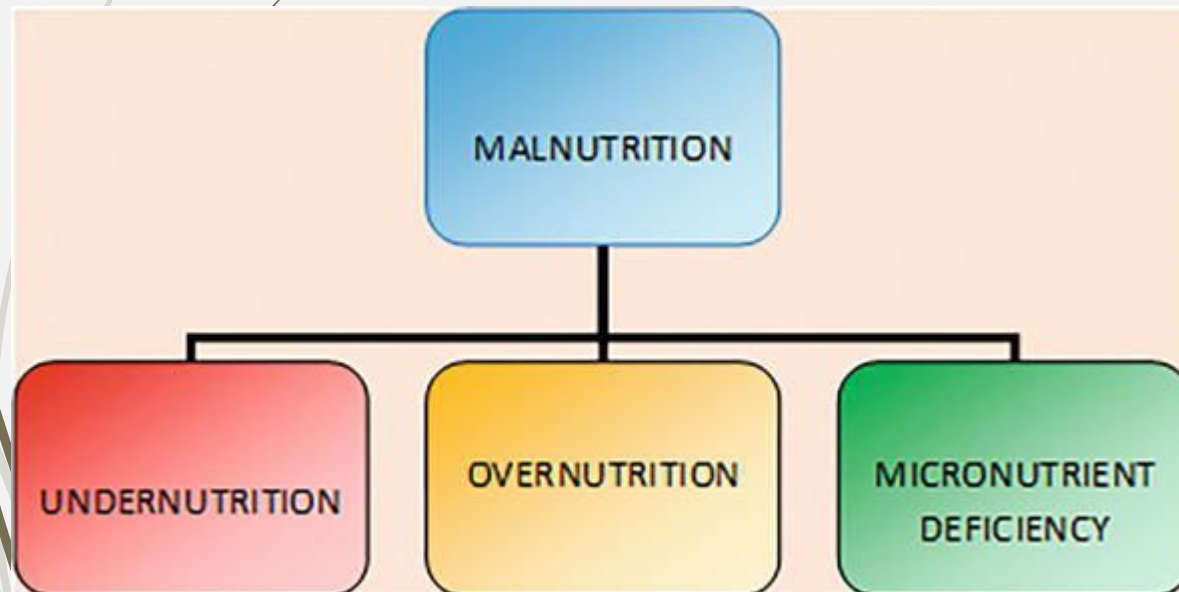
- ❑ **Malnutrition refers to:**
 - ❑ **“Deficiencies or excesses in nutrient intake, imbalance of essential nutrients or impaired nutrient utilization”. ‘WHO’**
 - ❑ **Malnutrition is a serious condition that happens when diet does not contain the right amount of nutrients. ‘NHS’**

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- ❑ It means "poor nutrition" and the double burden of malnutrition consists of both :
 - ❑ Undernutrition – not getting enough nutrients
 - ❑ Over nutrition – getting more nutrients than needed, overweight and obesity, as well as diet-related non communicable diseases.

CLASSIFICATION

It is officially classified by the World Health Organization into three broad categories:

- ❑ Undernutrition,
- ❑ Micronutrient-related malnutrition, and
- ❑ Over nutrition.



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- ❑ **Undernutrition, which includes wasting, stunting, underweight.**
- ❑ **Micronutrient-related malnutrition, which includes micronutrient deficiencies (a lack of important vitamins and minerals) or micronutrient excess; and**
- ❑ **Overweight, obesity and diet-related non communicable diseases (such as heart disease, stroke, diabetes and some cancers).**



UNDERNUTRITION

- ❑ Wasting (low weight-for-height).
- ❑ Stunting (low height-for-age).
- ❑ Underweight (low weight-for-age).



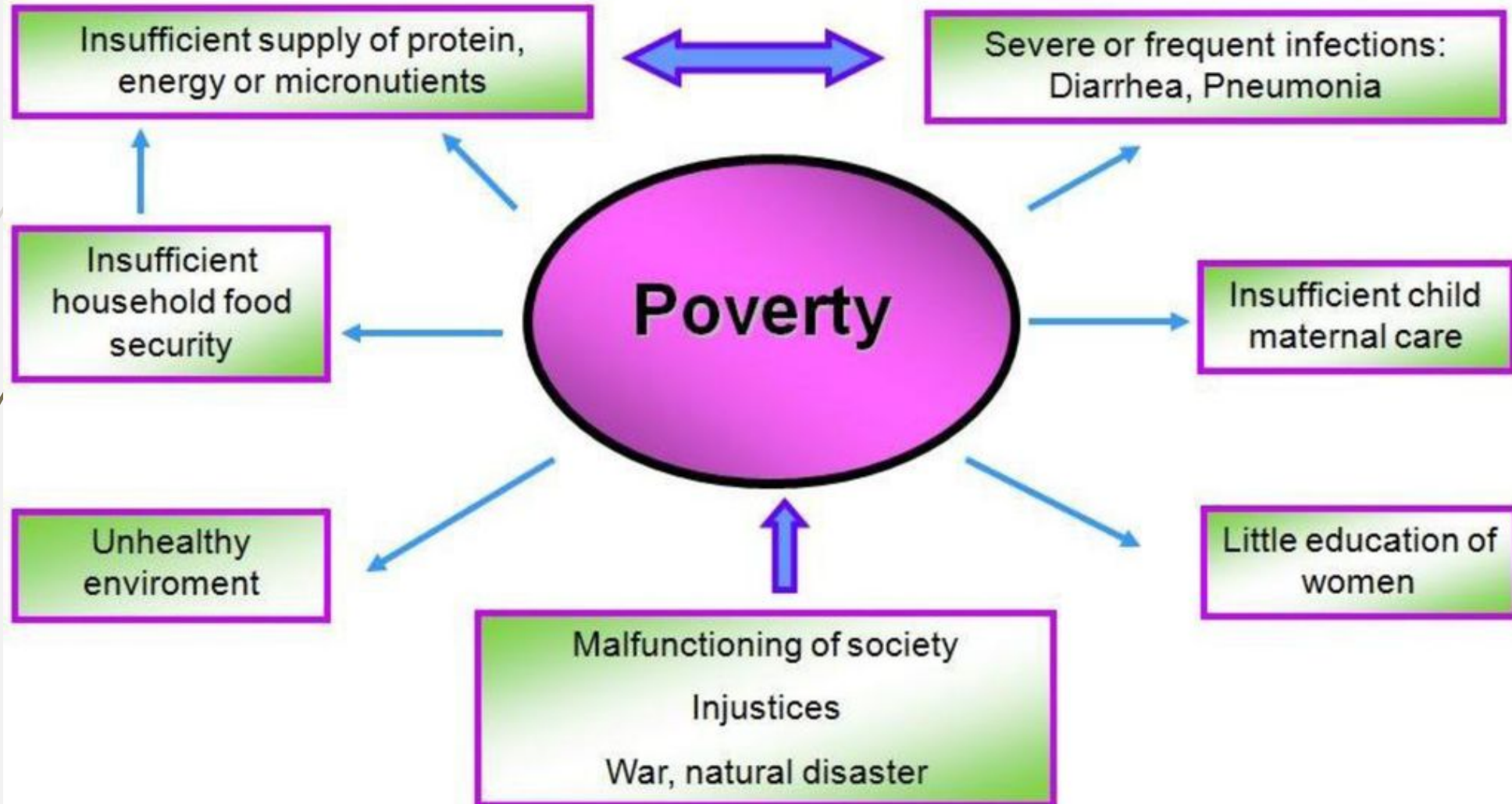
WHAT ARE THE CAUSES OF UNDERNUTRITION?

- ❑ Undernutrition is usually caused by not eating enough nutrients. It can also be caused by certain medical conditions that prevent your body from absorbing nutrients.
- ❑ Limited financial resources.
- ❑ Limited access to nutritious foods.
- ❑ Medical conditions that make eating difficult, such as nausea or difficulty swallowing.
- ❑ Medical conditions that deplete calories, such as chronic diarrhea or cancer.
- ❑ An extra need for calories, such as during pregnancy, breastfeeding or childhood.
- ❑ Mental health conditions that discourage eating, such as depression or dementia.
- ❑ Eating disorders such as anorexia and bulimia.
- ❑ Malabsorption disorders such as pancreatic insufficiency or inflammatory bowel disease.
- ❑ A very restricted diet or an unappealing diet chosen by someone else.

Malnutrition

Midle, moderate, severe (Marasmus-Kwashiorkor)

Micronutrient deficiencies



Causes of malnutrition

Altered nutrient processing

- Increased/altered metabolic demands
- Liver dysfunction

Inadequate intake

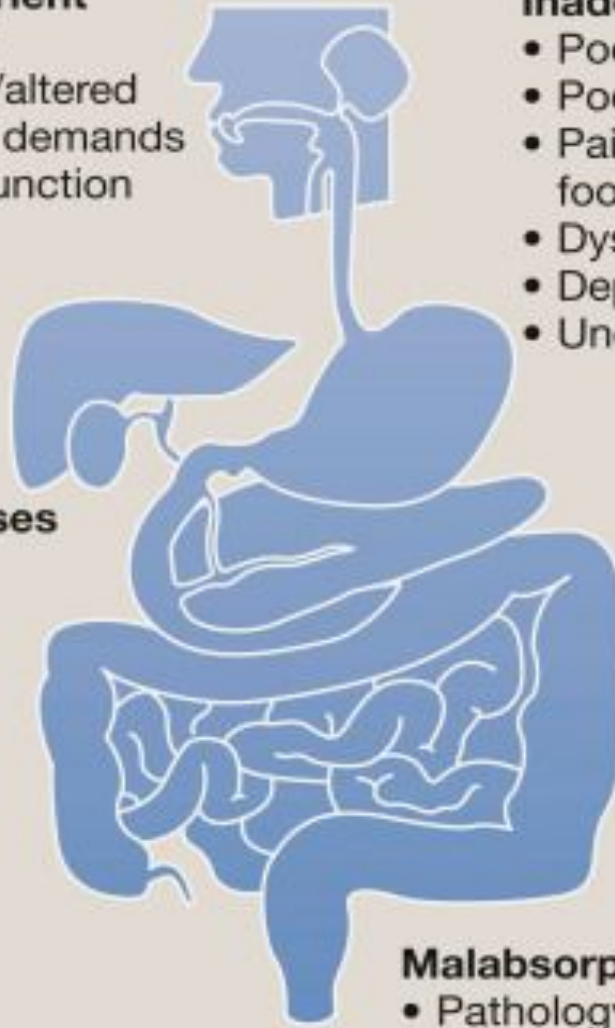
- Poor diet
- Poor appetite
- Pain/nausea with food
- Dysphagia
- Depression
- Unconsciousness

Excess losses

- Vomiting
- NG tube drainage
- Diarrhoea
- Surgical drains
- Fistulae
- Stomas

Malabsorption

- Pathology of stomach, intestine, pancreas and liver



NG, nasogastric

CAUSES OF OVER NUTRITION

- ❑ Few nutritious food options.
- ❑ A sedentary lifestyle.
- ❑ A condition that slows down the metabolism, such as hypothyroidism.
- ❑ A hormone imbalance that interferes with your hunger and fullness signals.
- ❑ Chronic stress.
- ❑ Anxiety or depression.
- ❑ Binge eating disorder.
- ❑ Chronic overuse of dietary supplements.



Populations more at risk of undernutrition

- ❑ Poor and low income
- ❑ Children
- ❑ Chronically ill
- ❑ Elderly



Populations more at risk of overnutrition

- ❑ Countries which have easier access to fast foods, which are high in calories but low in nutritional value, than they have to nutritious whole foods. This can lead to macronutrient over nutrition with micronutrient undernutrition.
 - ❑ Sedentary: Desk jobs, family obligations, health and social factors that keep people sitting all day instead of out and moving about can lead to significant weight gain.
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Disorders due to Malnutrition

1. Protein-energy malnutrition

- Kwashiorkor
- Marasmus

2. Under nutrition of vitamins and minerals

3. Obesity.

Effects of Nutrition on Oral Health

The nutritional needs for maintaining the health of the oral structures are similar to those for the reminder of the body.



Protein-Energy Malnutrition (PEM)

Marasmus, Kwashiorkor, and Marasmic Kwashiorkor

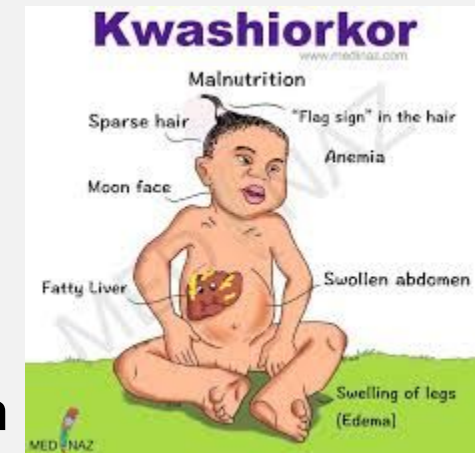
ORAL MANIFESTATIONS

- Bright reddening of tongue
- Loss of papillae: erythematous and smooth dorsum of tongue
- Kwashiorkor:
 - Edema of tongue with scalloping around the lateral margins due to indentation of the teeth.
- Bilateral angular cheilosis
- Fissuring of lip
- Loss of circumoral pigmentation
- Dry mouth
 - Reduced caries activity due to lack of substrate carbohydrate.
- Decreased overall growth of jaws
- Delayed eruption
- Deciduous teeth may show linear hypoplasia.



SPECIFIC ORAL MANIFESTATIONS WITH KWASHIORKOR

- ❑ Oral & Mucosal Ulcerations.
- ❑ Angular Cheilosis & Glossitis: cracked, bleeding lips (angular cheilosis) and a swollen, very red, or burning tongue (glossitis).
- ❑ High Risk of Dental Caries: Severe malnutrition reduces both the salivary flow rate and the buffering capacity of saliva. With the natural defense of saliva compromised, the risk of rampant tooth decay escalates significantly.
- ❑ Periodontal Complications: bone loss, weakened periodontal fibers, and hemorrhagic (bleeding) gingivitis.





Under nutrition of vitamins and minerals

Under nutrition of minerals

- **Calcium**
Rickets
- **Iodine deficiency**
Goiter
- **Iron deficiency**
Anemia
- **Zinc**
Growth retardation

Under nutrition of vitamins

- **Thiamine (Vitamin B1)**
Beriberi
- **Niacin (Vitamin B3)**
Pellagra
- **Vitamin C**
Scurvy
- **Vitamin D**
Rickets

Common Oral Manifestations by Deficiency

Understanding which symptoms align with specific nutrient gaps can help in identifying and addressing systemic malnutrition.

□ **Vitamin B Complex & Iron**: Deficiencies in B2, B6, B12, folic acid, and iron result in a burning sensation in the mouth (glossodynia), atrophic glossitis (a smooth, sore tongue), angular cheilitis (cracked, inflamed corners of the mouth), and oral ulcers.

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- **Vitamin C**: Essential for collagen formation, its deficiency causes capillary fragility, bleeding gums (scurvy), slow wound healing, and alveolar bone loss leading to loose teeth.
- **Vitamin A**: Crucial for epithelial health, a deficiency leads to salivary gland atrophy, dry mouth (xerostomia), and impaired tooth development.
- **Vitamin D & Calcium**: Lack of these nutrients causes enamel hypoplasia, pitting on the teeth surface, and an increased risk of periodontal disease.



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- **Zinc:** Zinc acts as an enzymatic cofactor required for cell growth and tissue repair. Oral manifestations includes the loss or alteration of taste due to papillary atrophy. It is also associated with dry mouth (xerostomia), gingivitis, and recurrent mucosal ulcerations.
- **Magnesium:** works alongside calcium to form and protect hard dental enamel. A lack of magnesium results in soft dental enamel unable to resist the acids that cause tooth decay. It is also associated with delayed tooth eruption and increased bleeding tendencies in gums.



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□ Fluoride:

- Fluoride is present in trace amounts in all mineralized tissues of the body such as enamel, dentin, and bone and plays an important role in the mineralization of bones and teeth, helps in tooth development, normal maintenance and hardening of dental enamel and prevention of “dental caries”.
 - Fluoride deficiency primarily compromises dental and skeletal health. It significantly weakens tooth enamel, directly leading to an increased rate of dental cavities, tooth decay, and heightened tooth sensitivity. Systemically, it can also contribute to weaker, more brittle bones and potentially accelerate conditions like osteoporosis.
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Common oral manifestations associated with overnutrition

- ❑ **Dental Caries (Cavities):** Diets high in refined carbohydrates and sugars promote the growth of acid-producing bacteria in the mouth. These bacteria attack tooth enamel, rapidly escalating the risk of severe tooth decay.
- ❑ **Gingivitis and Periodontitis:** Obesity triggers chronic, low-grade systemic inflammation and alters the body's immune response. This heightened inflammatory state accelerates the progression of gum disease and impedes the healing of periodontal tissues.



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- ❑ **Xerostomia (Dry Mouth):** Overnutrition can contribute to metabolic dysfunction, which is often linked to decreased salivary flow. Because saliva naturally washes away debris and neutralizes acids, this reduction significantly worsens cavity risks.
- ❑ **Enamel Defects:** In cases where overnutrition coexists with poor overall dietary quality, insufficient vitamin and mineral intake during childhood can lead to developmental defects such as enamel hypoplasia.
- ❑ **Candidal Infections:** High-sugar diets and metabolic conditions (like type 2 diabetes, which is a common consequence of overnutrition) create an environment that encourages the overgrowth of *Candida* species, leading to oral thrush.

Conclusion

- ❑ Oral health and malnutrition have a profound, two-way relationship. Deficiencies in vitamins and minerals directly cause structural defects and tissue inflammation in the mouth, while missing teeth or painful decay impair chewing and swallowing, significantly increasing the risk of systemic malnourishment.
- ❑ Proper nutrition balances the intake of calories, vitamins, and minerals and understanding how malnutrition affects oral structures is vital for preventing disease.





THANK YOU